

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc. I.T Semester V)
Applied Business Analytics

M2 Practical – IX

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9. Hypothesis Testing:

a. Formulate null and alternative hypotheses related to a business problem, conduct a hypothesis using appropriate statistical tests, and interpret the results.

Code:

```
import pandas as pd
import matplotlib.pyplot as plt
from scipy.stats import ttest_1samp
from statsmodels.stats.proportion import proportions_ztest
```

PART A: HYPOTHESIS TESTING

```
# Q1. Load the supermarket sales dataset
df = pd.read_csv("supermarket_sales.csv")
```

```
# Q2. Display first five records
print("\nQ2. First Five Records:")
print(df.head())
```

```
# Q3. Select revenue column
revenue = df["Total"]
```

```
print("\nQ3. Revenue Column:")
print(revenue)
```

```
# Q4. Check for missing values
print("\nQ4. Missing Values:")
print(revenue.isnull().sum())
```

```
# Q5. Calculate mean revenue
mean_revenue = revenue.mean()
print("\nQ5. Mean Revenue:")
print(mean_revenue)
```

```
# Q6. Expected average revenue
expected_revenue = 1500
print("\nQ6. Expected Average Revenue:")
print(expected_revenue)
```

```

# Q7. Null Hypothesis
print("\nQ7. Null Hypothesis:")
print("H0:  $\mu = 1500$ ")

# Q8. Alternative Hypothesis
print("\nQ8. Alternative Hypothesis:")
print("H1:  $\mu \neq 1500$ ")

# Q9. Significance level
alpha = 0.05
print("\nQ9. Significance Level:")
print(alpha)

# Q10. One-sample t-test
t_stat, p_value = ttest_1samp(revenue, expected_revenue)

print("\nQ10. One-Sample T-Test:")
print("T-statistic:", t_stat)
print("P-value:", p_value)

# Q11. P-value
print("\nQ11. P-value:")
print(p_value)

# Q12. Compare p-value with 0.05
print("\nQ12. Comparison:")
if p_value < alpha:
    print("P-value < 0.05")
else:
    print("P-value  $\geq$  0.05")

# Q13. Decision
print("\nQ13. Decision:")
if p_value < alpha:
    print("Reject the Null Hypothesis")
else:
    print("Do Not Reject the Null Hypothesis")

```

Output:

Q2. First Five Records:

	Invoice ID	Branch	City	Customer type	Gender	\
0	750-67-8428	A	Yangon	Member	Female	
1	226-31-3081	C	Naypyitaw	Normal	Female	
2	631-41-3108	A	Yangon	Normal	Male	
3	123-19-1176	A	Yangon	Member	Male	
4	373-73-7910	A	Yangon	Normal	Male	

	Product line	Unit price	Quantity	Tax 5%	Total	Date	\
0	Health and beauty	74.69	7	26.1415	548.9715	1/5/2019	
1	Electronic accessories	15.28	5	3.8200	80.2200	3/8/2019	
2	Home and lifestyle	46.33	7	16.2155	340.5255	3/3/2019	
3	Health and beauty	58.22	8	23.2880	489.0480	1/27/2019	
4	Sports and travel	86.31	7	30.2085	634.3785	2/8/2019	

	Time	Payment	cogs	gross margin percentage	gross income	Rating
0	13:08	Ewallet	522.83	4.761905	26.1415	9.1
1	10:29	Cash	76.40	4.761905	3.8200	9.6
2	13:23	Credit card	324.31	4.761905	16.2155	7.4
3	20:33	Ewallet	465.76	4.761905	23.2880	8.4
4	10:37	Ewallet	604.17	4.761905	30.2085	5.3

Q3. Revenue Column:

0	548.9715
1	80.2200
2	340.5255
3	489.0480
4	634.3785

995	42.3675
996	1022.4900
997	33.4320
998	69.1110
999	649.2990

Name: Total, Length: 1000, dtype: float64

Q4. Missing Values:

0

Q5. Mean Revenue:

322.966749

Q6. Expected Average Revenue:

1500

Q7. Null Hypothesis:

H0: $\mu = 1500$

Q8. Alternative Hypothesis:

H1: $\mu \neq 1500$

Q9. Significance Level:

0.05

Q10. One-Sample T-Test:

T-statistic: -151.37567896776733

P-value: 0.0

Q11. P-value:

0.0

Q12. Comparison:

P-value < 0.05

Q13. Decision:

Reject the Null Hypothesis

Conclusion:

The hypothesis test shows whether the average revenue is significantly different from ₹1,500. Based on the p-value and the 5% significance level, the appropriate conclusion is that the average revenue is significantly different from ₹1,500.

b. Perform A/B testing on a website or marketing campaign to evaluate the effectiveness of different strategies and make data-driven decisions

Code:

```
import pandas as pd
```

```

import matplotlib.pyplot as plt
from statsmodels.stats.proportion import proportions_ztest

# Q1. Load the A/B testing dataset
df = pd.read_csv("ab_testing.csv")

# Q2. Display the data
print("A/B Testing Data:")
print(df)

# Q3. Calculate conversion rate for Group A
group_a = df[df["Group"] == "A"].iloc[0]
rate_a = group_a["Conversions"] / group_a["Users"]

print("\nQ3. Conversion Rate for Group A:")
print(rate_a * 100, "%")

# Q4. Calculate conversion rate for Group B
group_b = df[df["Group"] == "B"].iloc[0]
rate_b = group_b["Conversions"] / group_b["Users"]

print("\nQ4. Conversion Rate for Group B:")
print(rate_b * 100, "%")

# Q5. Compare conversion rates
print("\nQ5. Comparison:")
print("Group A:", rate_a * 100, "%")
print("Group B:", rate_b * 100, "%")

# Q6. H0:  $p_A = p_B$ 

# Q7. H1:  $p_A \neq p_B$ 

# Q8. Significance level
alpha = 0.05

# Q9. Perform two-proportion test
conversions = [group_a["Conversions"], group_b["Conversions"]]
users = [group_a["Users"], group_b["Users"]]

z_stat, p_value = proportions_ztest(conversions, users)

print("\nQ9. Two-Proportion Test:")
print("Z-statistic:", z_stat)

# Q10. Calculate p-value
print("\nQ10. P-value:")
print(p_value)

```

```

# Q11. Compare p-value with 0.05
print("\nQ11. Comparison:")
if p_value < 0.05:
    print("P-value < 0.05")
else:
    print("P-value >= 0.05")

# Q12. Decision
print("\nQ12. Decision:")
if p_value < 0.05:
    print("Reject the Null Hypothesis")
else:
    print("Do Not Reject the Null Hypothesis")

# Q13. Create bar chart
groups = ["A", "B"]
conversion_rates = [rate_a * 100, rate_b * 100]

plt.bar(groups, conversion_rates)
plt.xlabel("Group")
plt.ylabel("Conversion Rate (%)")
plt.title("A/B Testing Conversion Rates")
plt.show()

```

Output:

```

A/B Testing Data:
  Group  Users  Conversions
0     A   1000         120
1     B   1000         150

Q3. Conversion Rate for Group A:
12.0 %

Q4. Conversion Rate for Group B:
15.0 %

Q5. Comparison:
Group A: 12.0 %
Group B: 15.0 %

Q9. Two-Proportion Test:
Z-statistic: -1.963049807622355

Q10. P-value:
0.04964038686536878

Q11. Comparison:
P-value < 0.05

Q12. Decision:
Reject the Null Hypothesis

```

